

STEPHENVILLE, NL, Canada

WMO: 718150

Lat: 48.5614N

Lon: 58.5661W

Elev: 81

StdP: 14.65

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.5	7.4	-3.8	4.5	4.9	0.9	5.8	8.5	41.5	23.1	36.7	22.1	11.4	280	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	74.1	65.0	71.7	63.8	69.7	63.2	67.2	71.5	65.8	69.8	64.5	68.3	9.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	94.3	69.7	64.1	89.9	68.5	62.8	85.8	67.3	31.8	71.7	30.7	69.9	29.7	68.3	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.0	26.5	22.7	DB	-2.2	80.5	4.8	2.7	-5.7	82.5	-8.5	84.0	-11.2	85.5	-14.7	87.5	(4)
(5)			WB	-2.9	70.6	4.8	2.1	-6.4	72.1	-9.2	73.3	-11.9	74.5	-15.3	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	22.7	21.6	26.5	35.9	45.4	53.8	61.9	62.8	55.4	46.0	37.5	29.1	(6)
(7)		DBStd	15.63	7.72	8.66	8.33	6.47	5.89	5.86	4.37	3.96	5.72	5.91	6.85	7.25	(7)
(8)		HDD50	4185	845	794	729	425	167	26	0	0	16	154	380	649	(8)
(9)		HDD65	8561	1310	1214	1194	874	606	338	116	89	291	589	826	1114	(9)
(10)		CDD50	1144	0	0	0	1	26	140	368	397	178	30	4	0	(10)
(11)		CDD65	45	0	0	0	0	0	2	19	21	3	0	0	0	(11)
(12)		CDH74	81	0	0	0	0	0	10	38	29	4	0	0	0	(12)
(13)		CDH80	4	0	0	0	0	0	1	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	10.9	13.8	13.3	12.1	11.4	9.9	8.6	8.0	8.6	9.6	10.9	11.9	13.1	(14)
(15)	Precipitation	PrecAvg	53.7	5.3	4.2	3.8	2.9	4.1	4.1	4.4	4.8	4.9	5.1	4.9	5.0	(15)
(16)		PrecMax	70.2	10.6	10.2	6.3	5.6	10.0	8.8	8.6	13.9	7.6	8.6	8.1	9.6	(16)
(17)		PrecMin	46.0	1.8	1.6	1.5	1.1	1.9	1.1	1.7	1.2	2.9	2.0	2.8	1.7	(17)
(18)		PrecStd	5.8	1.9	1.8	1.2	1.2	1.8	1.8	1.9	2.2	1.3	1.8	1.3	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	47.8	47.2	57.5	66.7	75.0	78.1	77.3	73.1	66.0	59.5	50.7	(19)
(20)			MCWB	45.4	45.4	43.2	50.2	54.1	61.4	66.4	66.0	65.4	60.7	56.3	48.4	(20)
(21)		2%	DB	41.6	41.8	43.6	52.7	62.1	69.8	74.2	73.9	69.1	61.1	55.3	46.1	(21)
(22)			MCWB	39.9	39.6	40.3	46.6	53.3	60.2	64.9	65.7	63.6	56.6	52.8	44.2	(22)
(23)		5%	DB	37.5	37.3	40.4	48.8	58.9	66.5	71.7	71.7	66.6	58.2	51.7	42.3	(23)
(24)			MCWB	35.8	34.9	37.3	43.7	51.9	58.8	63.5	64.5	61.8	54.3	49.5	40.3	(24)
(25)		10%	DB	34.0	33.8	37.5	45.6	55.8	63.6	69.3	69.9	64.6	55.4	47.7	39.0	(25)
(26)			MCWB	32.4	31.9	34.5	41.1	49.7	57.2	63.0	63.8	60.3	52.0	45.4	37.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	45.4	44.0	51.9	58.4	65.5	69.5	69.3	67.3	61.4	56.8	48.9	(27)
(28)			MCDWB	47.1	47.4	46.1	55.1	62.5	70.5	75.7	73.3	71.6	64.5	59.3	50.6	(28)
(29)		2%	WB	40.1	40.0	40.7	47.9	55.4	62.7	67.0	67.5	64.8	57.6	52.8	44.3	(29)
(30)			MCDWB	41.5	41.4	43.3	51.7	60.0	67.1	71.4	71.5	68.0	60.5	55.0	45.9	(30)
(31)		5%	WB	35.9	35.1	37.7	44.4	52.8	60.3	65.5	66.2	63.0	54.8	49.7	40.4	(31)
(32)			MCDWB	37.4	36.9	40.0	47.7	57.7	65.0	69.4	70.1	65.9	57.7	51.7	42.0	(32)
(33)		10%	WB	32.6	32.0	34.7	41.4	50.0	57.7	64.0	65.0	60.8	52.3	45.8	37.4	(33)
(34)			MCDWB	33.9	33.6	37.0	45.0	55.3	62.7	67.8	68.8	63.9	55.0	47.5	39.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.1	11.9	12.6	12.4	14.8	14.6	12.8	13.0	12.9	11.5	9.9	9.5	(35)
(36)			MCDBR	15.8	15.1	14.6	17.1	20.0	19.2	16.9	15.3	14.3	15.0	14.9	14.2	(36)
(37)			MCWBR	15.3	13.9	12.4	12.4	12.9	11.4	10.0	9.6	10.5	11.7	14.0	13.6	(37)
(38)		5% WB	MCDWB	16.2	15.2	14.4	16.2	17.1	16.3	12.5	12.6	12.8	13.5	14.9	14.2	(38)
(39)			MCWBR	15.8	14.4	13.1	13.2	12.5	11.3	8.6	9.2	10.6	11.7	14.2	13.8	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.304	0.301	0.326	0.351	0.365	0.376	0.411	0.396	0.370	0.343	0.316	0.298	(40)	
(42)		taud	2.314	2.359	2.351	2.387	2.422	2.417	2.359	2.400	2.453	2.478	2.468	2.378	(41)	
(43)		Ebn at Noon	238	271	282	285	285	281	268	268	264	253	236	226	(42)	
		Edh at Noon	23	27	32	35	35	36	37	34	30	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	223	421	869	1245	1617	1705	1632	1441	1064	654	317	176	(44)	
(45)		RadStd	25	43	93	128	111	161	135	75	82	45	35	19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.59	+0.67	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	+2.29	N/A	N/A	+0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page